

Propositional Logical Topology: Observation, Distinguishability, and Inference

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Abstract

Fix a finite propositional language and regard its truth assignments as points. A selected family of formulas acts as a family of observable tests. The topology generated by their truth regions records which logical states can be distinguished by those tests. We show that topological indistinguishability is equality of observation signatures, that the specialization preorder is inclusion of positive observations, and that a directed XOR-based quasi-pseudometric generates the same topology. Symmetrizing this directed distance gives weighted Hamming distance on observation signatures. We then separate two forms of information acquisition: refinement of the observable tests and restriction of the possible valuations by new premises. Classical inference becomes inclusion of the surviving truth region in the conclusion region, while open truth regions support the usual Heyting implication. The framework preserves the partition and valuation notation of propositional logical topology while distinguishing logical points, truth points, truth sets, topologies, and truth values.

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1 Introduction

Propositional formulas can be represented by the truth assignments on which they hold. If every such subset of a finite valuation space is declared open, the resulting topology is discrete and contains no structure beyond classical truth-set semantics. A nontrivial topological treatment begins instead with a restricted family of formulas that can be used as observations. Two truth assignments may then be logically different but observationally indistinguishable.

This paper develops five linked ideas:

- (i) complete truth assignments form a partition basis for propositional semantics;
- (ii) selected observable formulas determine distinguishability and indistinguishability;
- (iii) the specialization preorder records asymmetric logical separation;
- (iv) XOR of observation signatures gives symmetric and directed logical distances; and
- (v) adding observations refines the topology, while adding premises restricts the current space of possibilities.

The construction can be summarized as

$$\Phi \longmapsto o_\Phi \longmapsto \tau_\Phi \longleftrightarrow \preceq_\Phi \longleftrightarrow d_\Phi^+,$$

where Φ is a family of observable formulas, o_Φ is the observation signature, τ_Φ is the generated topology, $\preceq_\Phi$ is its specialization preorder, and d_Φ^+ is a directed logical distance.

The Boolean representation underlying the truth-region map is classical and is closely related to the representation of Boolean algebras by fields of sets [6]. Topological semantics for logical operations has a long history, notably in the work of McKinsey and Tarski [5]. Finite topologies and their order-theoretic structure are classical as well [2, 7]. Observation frames provide a related abstract separation between observable predicates and all predicates [3]. The purpose here is to assemble these ideas around the specific themes of partitioned propositional states, asymmetric distinguishability, XOR-based distance, refinement, and inference.

2 Valuations, truth regions, and partitions

Let $P = \{p_1, \dots, p_n\}$ be a finite set of propositional variables and let

$$\Omega_P = \{0, 1\}^P$$

be the set of valuations. For $v \in \Omega_P$, the expression $\mathcal{V}_v(\mathcal{L})$ is the value of a formula $\mathcal{L}$ at v .

2.1 Logical points, truth points, and truth sets

Definition 2.1 (Logical point and truth point). *A logical point of arity m is an ordered tuple of formulas*

$$\mathbf{L} = (\mathcal{L}_1, \dots, \mathcal{L}_m).$$

At a valuation $v \in \Omega_P$, its truth point is the Boolean vector

$$\mathcal{V}_v(\mathbf{L}) := (\mathcal{V}_v(\mathcal{L}_1), \dots, \mathcal{V}_v(\mathcal{L}_m)) \in \{0, 1\}^m.$$

A logical point is therefore syntactic, whereas a truth point is one of its possible semantic values. Neither should be confused with a valuation $v \in \Omega_P$, which assigns truth values to every variable in the fixed language and serves as a point of the topological carrier.

Definition 2.2 (Truth set or truth region). *For a formula $\mathcal{L}$, define*

$$\varsigma(\mathcal{L}) := \{v \in \Omega_P : \mathcal{V}_v(\mathcal{L}) = \mathbf{T}\}.$$

The notation $\mathcal{V}_v^T(\mathcal{L})$ abbreviates $v \in \varsigma(\mathcal{L})$, and $\mathcal{V}_v^F(\mathcal{L})$ abbreviates $v \notin \varsigma(\mathcal{L})$.

The original term *truth set* and the geometric term *truth region* refer to this same subset of Ω_P . We use “region” when emphasizing its role inside a topological space.

Using a fixed carrier ensures that ς depends only on the logical equivalence class of $\mathcal{L}$. It obeys

$$\begin{aligned} \varsigma(\mathcal{L}_1 \wedge \mathcal{L}_2) &= \varsigma(\mathcal{L}_1) \cap \varsigma(\mathcal{L}_2), \\ \varsigma(\mathcal{L}_1 \vee \mathcal{L}_2) &= \varsigma(\mathcal{L}_1) \cup \varsigma(\mathcal{L}_2), \\ \varsigma(\neg \mathcal{L}) &= \Omega_P \setminus \varsigma(\mathcal{L}). \end{aligned}$$

For two formulas, their union separates into an overlap and an exclusive remainder:

$$\varsigma(\mathcal{L}_1 \vee \mathcal{L}_2) = \varsigma(\mathcal{L}_1 \wedge \mathcal{L}_2) \dot{\cup} \varsigma(\mathcal{L}_1 \updownarrow \mathcal{L}_2).$$

Equivalently,

$$\mathcal{L}_1 \vee \mathcal{L}_2 \equiv (\mathcal{L}_1 \wedge \mathcal{L}_2) \updownarrow (\mathcal{L}_1 \updownarrow \mathcal{L}_2),$$

because the overlap and exclusive remainder are disjoint. The complete two-formula truth-pattern partition is

$$\begin{array}{ll} \mathcal{L}_1 \wedge \mathcal{L}_2, & \mathcal{L}_1 \wedge \neg \mathcal{L}_2, \\ \neg \mathcal{L}_1 \wedge \mathcal{L}_2, & \neg \mathcal{L}_1 \wedge \neg \mathcal{L}_2, \end{array}$$

after empty cells are removed. Thus XOR identifies the two exclusive cells; it is not a conversion of the conjunction into a disjunction.

For every $v \in \Omega_P$, define its complete minterm

$$m_v := \bigwedge_{i=1}^n \ell_i, \quad \ell_i = \begin{cases} p_i, & v(p_i) = 1, \\ \neg p_i, & v(p_i) = 0. \end{cases}$$

The family $\{\varsigma(m_v) : v \in \Omega_P\}$ is the singleton partition of Ω_P . Every formula has the canonical complete disjunctive form

$$\mathcal{L} \equiv \bigvee_{v \in \varsigma(\mathcal{L})} m_v.$$

Thus the manuscript's partitioned basis consists of the atoms of the finite Boolean algebra $\mathcal{P}(\Omega_P)$.

For two variables retain the notation

$$\begin{aligned} A &= X \wedge \neg Y, & B &= X \wedge Y, & \Omega_{X,Y} &= \{A, B, C, D\}. \\ C &= \neg X \wedge Y, & D &= \neg X \wedge \neg Y, \end{aligned}$$

For example,

$$\varsigma(X) = \{A, B\}, \quad \varsigma(Y) = \{B, C\}, \quad \varsigma(X \Rightarrow Y) = \{B, C, D\}.$$

3 Observation-generated ProLTs

The topology should record a limited observational vocabulary rather than all definable subsets.

Definition 3.1 (Observation family and signature). *Let*

$$\Phi = \{\varphi_1, \dots, \varphi_m\}$$

be a finite family of observable formulas. Put

$$b_i(v) := \mathbf{1}[v \in \varsigma(\varphi_i)]$$

and define the observation signature

$$o_\Phi(v) := (b_1(v), \dots, b_m(v)) \in \{0, 1\}^m.$$

Equivalently, we may regard $o_\Phi(v)$ as the index set

$$o_\Phi(v) = \{i : v \models \varphi_i\}.$$

After ordering Φ as the logical point $\varphi = (\varphi_1, \dots, \varphi_m)$, its truth point at v is exactly the observation signature:

$$\mathcal{V}_v(\varphi) = o_\Phi(v).$$

Thus the original truth-point language is retained, while the signature notation emphasizes that these coordinates are the observations available to the topology.

Definition 3.2 (Observation-generated ProLT). *The ProLT generated by Φ is*

$$\tau_\Phi := \text{Top}\{\varsigma(\varphi_i) : 1 \leq i \leq m\}$$

on the fixed carrier Ω_P . The displayed truth regions are a subbasis; their finite intersections form a basis.

We call (Ω_P, τ_Φ) the *truth topology generated by Φ* . Its points are valuations and its designated subbasic opens are truth sets. The carrier is always displayed because a family of subsets is not a topology until its underlying set is fixed.

The empty intersection of subbasic opens is Ω_P . Hence a convenient basis is

$$\mathcal{B}_\Phi = \left\{ \bigcap_{i \in I} \varsigma(\varphi_i) : I \subseteq \{1, \dots, m\} \right\}.$$

Order the Boolean cube $\{0, 1\}^m$ coordinatewise, or equivalently order its subsets by inclusion.

Theorem 3.3 (Signature representation). *The opens of τ_Φ are precisely the inverse images of upward-closed sets of realized observation signatures:*

$$\tau_\Phi = \{o_\Phi^{-1}(U) : U \subseteq o_\Phi(\Omega_P) \text{ is upward closed}\}.$$

Proof. For a realized signature s , its principal upward set is

$$\uparrow s = \{t : s \subseteq t\}.$$

Its inverse image is the basic open

$$o_\Phi^{-1}(\uparrow s) = \bigcap_{i \in s} \varsigma(\varphi_i).$$

Every upward-closed set in a finite poset is the union of the principal upward sets of its members. The result follows because τ_Φ consists of unions of its basic opens. $\square$

3.1 Partitions induced by observations

Define

$$v \sim_\Phi w \iff o_\Phi(v) = o_\Phi(w).$$

This equivalence relation partitions Ω_P into observational states.

Definition 3.4 (Partition topology). *For a partition Π of a set S , define*

$$\tau_\Pi := \left\{ \bigcup \mathcal{A} : \mathcal{A} \subseteq \Pi \right\}.$$

Its blocks are both open and closed.

An arbitrary observation topology need not be a partition topology. The difference is essential: partition topologies describe symmetric indistinguishability classes, whereas general finite topologies can also record one-way observational dependence.

Proposition 3.5 (Positive and two-sided observation). *Let*

$$\Phi^\pm = \Phi \cup \{\neg\varphi_i : 1 \leq i \leq m\}.$$

Then $\tau_{\Phi^\pm}$ is the partition topology whose blocks are the $\sim_\Phi$ -classes.

Proof. For every realized signature s , its fiber is

$$o_\Phi^{-1}(s) = \bigcap_{i \in s} \varsigma(\varphi_i) \cap \bigcap_{i \notin s} \varsigma(\neg\varphi_i).$$

It is therefore open in $\tau_{\Phi^\pm}$. Every open generated by $\Phi^\pm$ is saturated under equality of signatures, so it is a union of these fibers. $\square$

Example 3.6 (Two positive observations). *Take $\Phi = \{X, Y\}$ on $\Omega_{X,Y}$. Then*

$$\tau_\Phi = \{\emptyset, \{B\}, \{A, B\}, \{B, C\}, \{A, B, C\}, \Omega_{X,Y}\}.$$

Adding $\neg X$ and $\neg Y$ makes each of A, B, C, D open and gives the discrete topology.

4 Distinguishability and separation

Three questions should not be conflated:

1. Formulas $\mathcal{L}_1, \mathcal{L}_2$ are semantically different when $\varsigma(\mathcal{L}_1) \neq \varsigma(\mathcal{L}_2)$.
2. Valuations v, w are observationally distinguishable by Φ when $o_\Phi(v) \neq o_\Phi(w)$.
3. Points are topologically separated according to the neighborhoods available in τ_Φ .

Overlap between two truth regions does not by itself establish topological indistinguishability.

Proposition 4.1 (Semantic XOR criterion). *For formulas $\mathcal{L}_1, \mathcal{L}_2$,*

$$\varsigma(\mathcal{L}_1) \neq \varsigma(\mathcal{L}_2) \iff \varsigma(\mathcal{L}_1 \updownarrow \mathcal{L}_2) \neq \emptyset.$$

At a particular valuation v , their truth values are distinct exactly when $v \in \varsigma(\mathcal{L}_1 \updownarrow \mathcal{L}_2)$.

Proof. The XOR truth region is the symmetric difference

$$\varsigma(\mathcal{L}_1) \triangle \varsigma(\mathcal{L}_2).$$

It is nonempty exactly when the two truth regions are unequal. □

Observing only $\mathcal{L}_1 \wedge \mathcal{L}_2$ gives one positive bit and groups together all three ways in which the conjunction can be false. Replacing that observation by the pair $\Phi = \{\mathcal{L}_1, \mathcal{L}_2\}$ refines those states according to the realized truth points $(1, 1), (1, 0), (0, 1), (0, 0)$. Adding both negations makes the nonempty truth-pattern cells clopen. This is the precise observational sense in which the conjuncts become distinguishable.

Definition 4.2 (Topological indistinguishability). *Points $v, w \in \Omega_P$ are topologically indistinguishable when they belong to exactly the same open sets of τ_Φ .*

Theorem 4.3 (Indistinguishability theorem). *For $v, w \in \Omega_P$, the following are equivalent:*

- (i) *v and w are topologically indistinguishable in τ_Φ ;*
- (ii) *v and w have the same subbasic open memberships;*
- (iii) *$o_\Phi(v) = o_\Phi(w)$;*
- (iv) *$v \sim_\Phi w$.*

Proof. Equality of all open memberships implies equality of subbasic memberships. Subbasic memberships are exactly the coordinates of o_Φ . Conversely, equality of signatures gives equality of membership in every finite intersection of subbasic opens and hence in every union of such intersections. □

Corollary 4.4 (T_0 quotient). *The space (Ω_P, τ_Φ) is T_0 exactly when o_Φ is injective. Its Kolmogorov quotient is naturally identified with the set $o_\Phi(\Omega_P)$ of realized signatures.*

The next separation level requires distinguishability in both directions.

Proposition 4.5 (T_1 and complete finite separation). *The observation space is T_1 exactly when, for every distinct v, w ,*

$$o_\Phi(v) \not\subseteq o_\Phi(w) \quad \text{and} \quad o_\Phi(w) \not\subseteq o_\Phi(v).$$

Equivalently, the realized signatures form an antichain and are distinct. A finite T_1 space is discrete; therefore finite Hausdorff observation spaces are discrete as well.

Proof. There is an open containing v but not w exactly when some observation is true at v and false at w , which is equivalent to $o_\Phi(v) \not\subseteq o_\Phi(w)$. The T_1 condition requires this in both directions. In a finite T_1 space, each singleton is the finite intersection of open complements of the other singletons, so every singleton is open. $\square$

Example 4.6 (Sierpiński observation space). *Let $P = \{X\}$ and $\Phi = \{X\}$. With valuations $0, 1$, the topology is*

$$\tau_\Phi = \{\emptyset, \{1\}, \{0, 1\}\}.$$

It is T_0 but not T_1 . The positive observation X distinguishes 1 from 0, but no positive observation in Φ distinguishes 0 from 1. This is the canonical two-point example of asymmetric observation.

5 Asymmetric separation and the specialization preorder

We use the specialization convention

$$v \preceq_\Phi w \iff \text{every open neighborhood of } v \text{ contains } w.$$

Some authors use the reverse convention; the direction is fixed here by the display above.

Because Ω_P is finite, every point has a smallest open neighborhood

$$N_\Phi(v) := \bigcap \{U \in \tau_\Phi : v \in U\}.$$

Theorem 5.1 (Positive-information order). *For all $v, w \in \Omega_P$,*

$$v \preceq_\Phi w \iff o_\Phi(v) \subseteq o_\Phi(w).$$

Moreover,

$$N_\Phi(v) = \bigcap_{i \in o_\Phi(v)} \varsigma(\varphi_i) = \{w : o_\Phi(v) \subseteq o_\Phi(w)\}.$$

Proof. If every open containing v contains w , this holds for every subbasic open, so every observation true at v is true at w . Conversely, if $o_\Phi(v) \subseteq o_\Phi(w)$, then every basic open containing v contains w , and therefore every open containing v contains w . The neighborhood formula follows. $\square$

Thus w preserves all positive information observable at v and may satisfy additional observations. Failure of symmetry is not a defect: it records that positive evidence can separate one direction without separating the reverse. After quotienting by $\sim_\Phi$, the preorder becomes the inclusion order on realized signatures.

6 XOR difference and logical distance

The original XOR construction naturally produces a difference vector. Let $a_1, \dots, a_m$ be positive weights.

Definition 6.1 (Observable XOR difference). *Define*

$$\Delta_\Phi(v, w) := o_\Phi(v) \updownarrow o_\Phi(w) := (b_i(v) \updownarrow b_i(w))_{i=1}^m.$$

This vector identifies exactly which observations disagree.

Definition 6.2 (Symmetric logical distance). *Define the weighted Hamming distance between signatures by*

$$d_\Phi(v, w) := \sum_{i=1}^m a_i |b_i(v) - b_i(w)|.$$

Proposition 6.3. *The function d_Φ is a pseudometric on Ω_P , and*

$$d_\Phi(v, w) = 0 \iff v \sim_\Phi w.$$

Consequently, d_Φ induces a metric on the T_0 quotient $\Omega_P / \sim_\Phi$.

Proof. Each coordinate contributes the weighted discrete metric on $\{0, 1\}$; their sum is a pseudometric. It vanishes exactly when every observation coordinate agrees. $\square$

Symmetric distance measures disagreement but cannot by itself encode the direction of the specialization preorder. For that purpose define the cost of losing observations.

Definition 6.4 (Directed logical distance). *Define*

$$d_\Phi^+(v, w) := \sum_{i=1}^m a_i \mathbf{1}[b_i(v) = 1 \text{ and } b_i(w) = 0].$$

Theorem 6.5 (Quasi-metric representation). *The function d_Φ^+ is a quasi-pseudometric: it is nonnegative, $d_\Phi^+(v, v) = 0$, and*

$$d_\Phi^+(v, u) \leq d_\Phi^+(v, w) + d_\Phi^+(w, u).$$

It also satisfies

$$d_\Phi^+(v, w) = 0 \iff v \preceq_\Phi w,$$

and

$$d_\Phi(v, w) = d_\Phi^+(v, w) + d_\Phi^+(w, v).$$

The topology generated by the forward balls

$$B^+(v, r) := \{w : d_\Phi^+(v, w) < r\}$$

is exactly τ_Φ .

Proof. For each coordinate, an observation lost from v to u must be lost either from v to w or from w to u . Summing the resulting coordinatewise inequality proves the triangle inequality. Vanishing means that no observation true at v is false at w , which is exactly $o_\Phi(v) \subseteq o_\Phi(w)$. The symmetrization identity follows because every disagreement is a loss in exactly one direction.

Every forward ball is upward closed in $\preceq_\Phi$: if w belongs to the ball and $w \preceq_\Phi u$, then u loses no more observations from v than w does. Hence every forward ball is τ_Φ -open. If $0 < r \leq \min_i a_i$, then

$$B^+(v, r) = N_\Phi(v).$$

The smallest neighborhoods generate τ_Φ , so the forward-ball topology is exactly τ_Φ . $\square$

This result makes the topology, asymmetric separation, and directed distance three descriptions of the same finite information structure. It is compatible with the general role of nonsymmetric distance in generalized metric theory [4].

7 Interior, closure, and observational boundary

Topological operators describe what can be verified or ruled out using the chosen observations.

Definition 7.1. *For an event $E \subseteq \Omega_P$, let*

$$\text{Int}_\Phi(E) := \text{Int}_{\tau_\Phi}(E), \quad \text{Cl}_\Phi(E) := \text{Cl}_{\tau_\Phi}(E), \quad \partial_\Phi E := \text{Cl}_\Phi(E) \setminus \text{Int}_\Phi(E).$$

The interior $\text{Int}_\Phi(E)$ is the largest observable open event contained in E . If the actual valuation lies in this interior, some available positive observation certifies membership in E . The closure $\text{Cl}_\Phi(E)$ consists of valuations at which the available observations cannot exclude contact with E . The boundary $\partial_\Phi E$ is therefore an observational ambiguity region. Equivalently,

$$\partial_\Phi E = \text{Cl}_\Phi(E) \cap \text{Cl}_\Phi(\Omega_P \setminus E).$$

It contains precisely those points whose available neighborhoods cannot settle membership in E one way or the other.

Proposition 7.2 (Finite neighborhood formulas). *For every $E \subseteq \Omega_P$,*

$$\begin{aligned} \text{Int}_\Phi(E) &= \{v : N_\Phi(v) \subseteq E\}, \\ \text{Cl}_\Phi(E) &= \{v : N_\Phi(v) \cap E \neq \emptyset\}. \end{aligned}$$

Consequently,

$$v \in \partial_\Phi E \iff N_\Phi(v) \cap E \neq \emptyset \text{ and } N_\Phi(v) \cap (\Omega_P \setminus E) \neq \emptyset.$$

Proof. The point v is interior to E exactly when it has an open neighborhood contained in E . Since $N_\Phi(v)$ is its smallest neighborhood, this is equivalent to $N_\Phi(v) \subseteq E$. A point belongs to the closure of E exactly when each of its neighborhoods meets E , which in a finite space is equivalent to $N_\Phi(v) \cap E \neq \emptyset$. $\square$

Remark 7.3 (Boundary versus XOR). *The symbol ∂_Φ denotes the topological boundary operator above. It is not the “unfolding” operation that removes the overlap of two formulas. That Boolean operation is the exclusive remainder*

$$\varsigma(\mathcal{L}_1 \vee \mathcal{L}_2) \setminus \varsigma(\mathcal{L}_1 \wedge \mathcal{L}_2) = \varsigma(\mathcal{L}_1 \updownarrow \mathcal{L}_2).$$

The two constructions can differ even in the simplest cases: in a discrete truth topology every topological boundary is empty, while an XOR truth region may be nonempty.

Example 7.4 (Boundary of implication under positive observations). For $\Phi = \{X, Y\}$,

$$E = \varsigma(X \Rightarrow Y) = \{B, C, D\}.$$

Using the topology in Example 3.5,

$$\text{Int}_\Phi(E) = \{B, C\}, \quad \text{Cl}_\Phi(E) = \Omega_{X,Y}, \quad \partial_\Phi E = \{A, D\}.$$

Thus $X \Rightarrow Y$ is neither open nor closed in this observation topology. The point D satisfies the implication but is not positively certifiable as such using only X and Y , while A lies observationally adjacent to satisfying states despite falsifying the implication.

Example 7.5 (XOR region under coarse and discrete observation). Retain the partition labels $A = X \wedge \neg Y$, $B = X \wedge Y$, $C = \neg X \wedge Y$, and $D = \neg X \wedge \neg Y$. The exclusive truth region is

$$E_\oplus := \varsigma(X \updownarrow Y) = \{A, C\}.$$

First use only the positive observations $\Phi = \{X, Y\}$. The smallest neighborhoods are

$$N_\Phi(A) = \{A, B\}, \quad N_\Phi(B) = \{B\}, \quad N_\Phi(C) = \{B, C\}, \quad N_\Phi(D) = \Omega_{X,Y}.$$

Hence no nonempty τ_Φ -open set is contained in $E_\oplus$, while the neighborhoods of A , C , and D each meet it. Thus

$$\text{Int}_\Phi(E_\oplus) = \emptyset, \quad \text{Cl}_\Phi(E_\oplus) = \{A, C, D\}, \quad \partial_\Phi E_\oplus = \{A, C, D\}.$$

The points A and C belong to the XOR region but are not positively certifiable using only X and Y , because their smallest neighborhoods also contain B . The point D is not in the XOR region, but it remains on its boundary because every positive-observation neighborhood of D contains both XOR and non-XOR valuations.

Now take the two-sided observation family

$$\Phi^\pm = \{X, \neg X, Y, \neg Y\}.$$

Its topology is discrete. Therefore $E_\oplus$ is both open and closed, and

$$\text{Int}_{\Phi^\pm}(E_\oplus) = \text{Cl}_{\Phi^\pm}(E_\oplus) = \{A, C\}, \quad \partial_{\Phi^\pm} E_\oplus = \emptyset.$$

The XOR truth set is unchanged by the topology; only its observable boundary changes. This is the distinction between Boolean unfolding and topological ambiguity. Figure 1 displays the same calculation on the four truth-pattern cells.

7.1 Open truth regions and Heyting implication

The opens of any topology form a Heyting algebra. For $U, V \in \tau_\Phi$, define

$$U \Rightarrow_H V := \text{Int}_\Phi((\Omega_P \setminus U) \cup V), \quad \neg_H U := \text{Int}_\Phi(\Omega_P \setminus U).$$

Then

$$U \cap (U \Rightarrow_H V) \subseteq V$$

is topological Modus Ponens, and

$$(U \Rightarrow_H V) \cap \neg_H V \subseteq \neg_H U$$

is topological Modus Tollens. In a discrete topology the interiors do nothing, and these reduce to their classical truth-set forms. In a nondiscrete observation topology, the interior records the loss imposed by requiring the result to remain observable.

	X	$\neg X$
Y	$B = X \wedge Y$ non-XOR	$C = \neg X \wedge Y$ XOR
$\neg Y$	$A = X \wedge \neg Y$ XOR	$D = \neg X \wedge \neg Y$ non-XOR

Figure 1: The four truth-pattern cells for X and Y . The bold cells form $E_{\oplus} = \varsigma(X \updownarrow Y) = \{A, C\}$. Under positive observation $\Phi = \{X, Y\}$, the observable boundary is $\partial_{\Phi} E_{\oplus} = \{A, C, D\}$; under $\Phi^{\pm}$ it is empty.

8 Information refinement and logical update

The phrase “acquiring information” can describe two mathematically different operations.

8.1 Refining the observations

Suppose $\Phi \subseteq \Psi$. Then Ψ can make at least all distinctions available to Φ .

Theorem 8.1 (Refinement theorem). *If $\Phi \subseteq \Psi$, then:*

- (i) $\tau_{\Phi} \subseteq \tau_{\Psi}$;
- (ii) $v \sim_{\Psi} w$ implies $v \sim_{\Phi} w$;
- (iii) $\preceq_{\Psi} \subseteq \preceq_{\Phi}$ as relations;
- (iv) if the weights on Φ are retained, then $d_{\Psi}^{+}(v, w) \geq d_{\Phi}^{+}(v, w)$ and $d_{\Psi}(v, w) \geq d_{\Phi}(v, w)$.

Proof. The subbasis for τ_{Φ} is contained in the subbasis for τ_{Ψ} , proving (i). Equality of the longer signatures implies equality after deleting the new coordinates, proving (ii). Inclusion of Ψ -signatures implies inclusion after deleting those coordinates, proving (iii). The added coordinates make nonnegative contributions to both distances, proving (iv). $\square$

Adding an observable can split an indistinguishability block, remove a one-way comparison, and increase measured separation. A proposed observation ψ is topologically redundant relative to Φ exactly when

$$\varsigma(\psi) \in \tau_{\Phi},$$

because adding an already open truth region does not refine the topology.

8.2 Updating the possible valuations

Let $S \subseteq \Omega_P$ be the current possibility region. Learning a premise γ produces

$$S \longmapsto S \cap \varsigma(\gamma).$$

For premises $\Gamma = \{\gamma_1, \dots, \gamma_r\}$, define

$$S_{\Gamma} := \bigcap_{j=1}^r \varsigma(\gamma_j).$$

The updated observation space is the subspace

$$\mathbf{X}_\Gamma^T := (S_\Gamma, \tau_\Phi|_{S_\Gamma}), \quad \tau_\Phi|_S := \{U \cap S : U \in \tau_\Phi\}.$$

Sequential updates are associative, commutative, and idempotent because set intersection has those properties. Inconsistent premises produce the empty space.

Observation refinement and premise update need not have the same effect. Refinement changes which distinctions are expressible; update changes which worlds remain possible.

9 Inference by truth-region restriction

Classical inference is a relationship between the common truth region of the premises and the truth region of the conclusion.

Theorem 9.1 (Semantic consequence). *For a finite set of premises Γ and a conclusion ψ ,*

$$\Gamma \models \psi \iff S_\Gamma \subseteq \varsigma(\psi).$$

When this holds, the inclusion

$$\mathbf{X}_\Gamma^T \hookrightarrow (\varsigma(\psi), \tau_\Phi|_{\varsigma(\psi)})$$

is continuous.

Proof. The semantic statement says exactly that every valuation satisfying all premises satisfies ψ . The inclusion between two subspaces of the same ambient space is continuous. $\square$

9.1 Minimal truth ProLTs

To retain the original formula-indexed ProLT notation, define

$$\mathcal{T}_S^T := (S, \{\emptyset, S\})$$

and call this the *minimal truth topology on S* . It is distinct from the ambient truth topology (Ω_P, τ_Φ) and from the subspace topology $\tau_\Phi|_S$: its carrier is S and its only opens are $\emptyset$ and S . Keeping the carrier in the ordered pair prevents these three constructions from being conflated. We then define a carrier-induced conjunction by

$$\mathcal{T}_R^T \hat{\cap} \mathcal{T}_S^T := \mathcal{T}_{R \cap S}^T.$$

The symbol $\hat{\cap}$ is not ordinary intersection of the two families of open sets. It means: intersect the carriers, then equip the result with its minimal truth topology. Order these spaces by

$$\mathcal{T}_R^T \preceq \mathcal{T}_S^T \iff R \subseteq S.$$

Then $\hat{\cap}$ is associative, commutative, and idempotent, with unit $\mathcal{T}_{\Omega_P}^T$ and zero $\mathcal{T}_\emptyset^T$; it is the greatest lower bound under $\preceq$.

9.2 Modus Ponens and Modus Tollens

Using A, B, C, D ,

$$\begin{aligned} \varsigma(X \Rightarrow Y) \cap \varsigma(X) &= \{B\} = \varsigma(X \wedge Y) \subseteq \varsigma(Y), \\ \varsigma(X \Rightarrow Y) \cap \varsigma(\neg Y) &= \{D\} = \varsigma(\neg X \wedge \neg Y) \subseteq \varsigma(\neg X). \end{aligned}$$

Therefore

$$\begin{aligned} \mathcal{T}_{\varsigma(X \Rightarrow Y)}^T \hat{\cap} \mathcal{T}_{\varsigma(X)}^T &\preceq \mathcal{T}_{\varsigma(Y)}^T, \\ \mathcal{T}_{\varsigma(X \Rightarrow Y)}^T \hat{\cap} \mathcal{T}_{\varsigma(\neg Y)}^T &\preceq \mathcal{T}_{\varsigma(\neg X)}^T. \end{aligned}$$

9.3 Hypothetical Syllogism

For three variables retain the partition labels

$$\begin{aligned} A &= X \wedge \neg Y \wedge \neg Z, & B &= \neg X \wedge Y \wedge \neg Z, & C &= \neg X \wedge \neg Y \wedge Z, & D &= X \wedge Y \wedge \neg Z, \\ E &= X \wedge \neg Y \wedge Z, & F &= \neg X \wedge Y \wedge Z, & G &= X \wedge Y \wedge Z, & H &= \neg X \wedge \neg Y \wedge \neg Z. \end{aligned}$$

Then

$$\begin{aligned} \varsigma(X \Rightarrow Y) &= \{B, C, D, F, G, H\}, \\ \varsigma(Y \Rightarrow Z) &= \{A, C, E, F, G, H\}, \\ \varsigma(X \Rightarrow Z) &= \{B, C, E, F, G, H\}. \end{aligned}$$

Hence

$$\varsigma(X \Rightarrow Y) \cap \varsigma(Y \Rightarrow Z) = \{C, F, G, H\} \subsetneq \varsigma(X \Rightarrow Z).$$

The proper inclusion is the correct statement of validity. Equality is not required because the premises contain more information than the conclusion. Figure 2 makes the truth-region calculation explicit: the boxed cells satisfy both premises and every one satisfies the conclusion.

cell	$X \Rightarrow Y$	$Y \Rightarrow Z$	both premises	$X \Rightarrow Z$
A	F	T	F	T
B	T	F	F	T
C	T	T	T	T
D	T	F	F	F
E	F	T	F	T
F	T	T	T	T
G	T	T	T	T
H	T	T	T	T

Figure 2: Hypothetical syllogism on the eight three-variable truth-pattern cells. The boxed entries are $\varsigma(X \Rightarrow Y) \cap \varsigma(Y \Rightarrow Z) = \{C, F, G, H\}$; they all lie in $\varsigma(X \Rightarrow Z)$. The operation is an intersection of truth regions, equivalently the pullback of their truth subspaces, not an intersection of the families of open sets.

9.4 Pullback formulation

For a single formula $\mathcal{L}$, write

$$\mathbf{X}_{\mathcal{L}}^T := (\varsigma(\mathcal{L}), \tau_{\Phi}|_{\varsigma(\mathcal{L})}).$$

The inclusions of two truth subspaces into (Ω_P, τ_{Φ}) have pullback

$$\mathbf{X}_{\mathcal{L}_1}^T \times_{(\Omega_P, \tau_{\Phi})} \mathbf{X}_{\mathcal{L}_2}^T \cong \mathbf{X}_{\mathcal{L}_1 \wedge \mathcal{L}_2}^T.$$

For Modus Ponens,

$$\mathbf{X}_{X \Rightarrow Y}^T \times_{(\Omega_{X,Y}, \tau_{\Phi})} \mathbf{X}_X^T \cong \mathbf{X}_{X \wedge Y}^T \hookrightarrow \mathbf{X}_Y^T.$$

This is the standard topological construction closest to the idea of forming a relative space from several premises.

10 Continuous maps and preservation of information

Let (Ω_P, τ_Φ) and (Ω_Q, τ_Ψ) be observation spaces. A map

$$f : \Omega_P \longrightarrow \Omega_Q$$

is continuous when the inverse image of every Ψ -observable open event is Φ -observable. It suffices to check the subbasic truth regions:

$$f^{-1}(\varsigma(\psi)) \in \tau_\Phi \quad (\psi \in \Psi).$$

Proposition 10.1 (Monotonicity of continuous maps). *If f is continuous and $v \preceq_\Phi w$, then*

$$f(v) \preceq_\Psi f(w).$$

Proof. Let $U \in \tau_\Psi$ contain $f(v)$. Then $f^{-1}(U)$ is a Φ -open set containing v , so it contains w . Hence U contains $f(w)$. $\square$

Continuity therefore preserves one-way observational dependence. This gives a rigorous starting point for the manuscript's proposed logical morphisms.

11 Scope, limitations, and deferred directions

The results above form a coherent finite theory of observation, distinguishability, distance, refinement, and inference. The following ideas from the original research notes are retained as explicit bridges to companion papers rather than asserted as results here.

11.1 Logical complexes and homology

A natural future construction is the nerve of a family of truth regions. For an observation family Φ , define a simplicial complex

$$K_\Phi := \left\{ \sigma \subseteq \Phi : \bigcap_{\varphi \in \sigma} \varsigma(\varphi) \neq \emptyset \right\}.$$

Its simplices are jointly satisfiable families of observations. Homology of K_Φ could then measure higher-order patterns of compatibility not visible from pairwise intersections alone. This construction, its invariance, and its interpretation are reserved for the separate logical-homology paper.

11.2 Sheaves and contextuality

A sheaf-theoretic extension would require a cover by measurement or reasoning contexts, sets of local valuations on each context, and restriction maps on overlaps. The central question would be whether compatible local sections glue to a global valuation. This is structurally related to established sheaf-theoretic contextuality [1] and requires a separate treatment.

11.3 Cognitive interpretation

The contrast between coarse and fine observation topologies may eventually model limited versus refined conceptual discrimination. The formal content available here is the splitting of indistinguishability classes and the removal of specialization comparisons under refinement. Claims about consciousness, neural organization, or psychological impairment require an independently specified empirical model and are therefore deferred.

11.4 Quantum interpretation

Topological indistinguishability is not by itself quantum entanglement. A quantum application would need measurement contexts, compatibility relations, probability assignments, and a precise nonclassical obstruction. The logical complex and sheaf directions above provide possible rigorous entry points, but no quantum claim is made in the present paper.

11.5 Filters, limits, and infinite languages

Filters and convergence become potentially useful when the language, the valuation space, or a chain of observational refinements is infinite. In the present finite setting they add little beyond smallest neighborhoods and the specialization preorder. They are reserved for an extension involving asymptotic information acquisition.

12 Conclusion

A propositional logical topology becomes nontrivial when it records a selected family of observations rather than every definable truth set. Observation signatures simultaneously determine a partition into indistinguishable states, a finite topology, and its specialization preorder. Coordinatewise XOR records which observations differ; weighted Hamming distance measures symmetric disagreement; and the directed distance d_{Φ}^+ measures loss of positive information while generating the original observation topology. Adding tests refines distinguishability, while adding premises restricts the possibility space. Inference is the inclusion of that restricted region in the truth region of a conclusion.

The framework gives precise mathematical forms to partitioning, distinguishability, asymmetric separation, logical distance, and refinement. Its individual ingredients are classical; a further research contribution should build on them through a new algorithm, complexity result, or validated application.

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